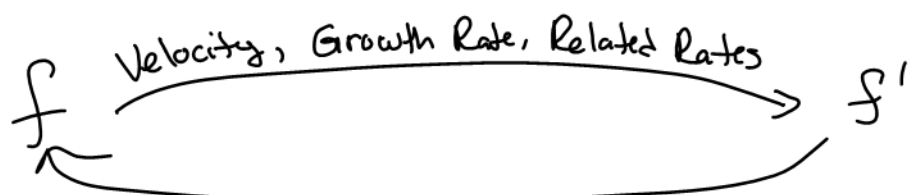


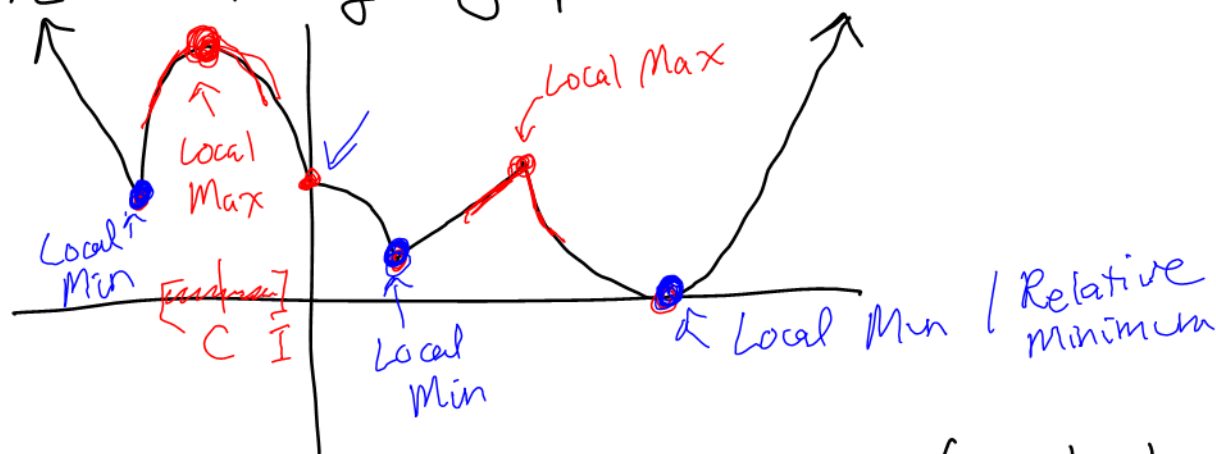
Lecture 16: Relative Extrema and Critical Numbers

Goal: Discuss when a function's behavior changes.



Extrema, Intervals of Inc./Dec., Concavity, Inflection Points

Consider the following graph:

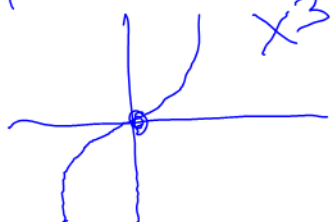


Def A fun f has a relative maximum (or local max) at $x=c$, if there is an interval I such that $f(x) \leq f(c)$ for $x \in I$.
 (Contains c)

Def To obtain local mins, replace " $f(x) \leq f(c)$ " with " $f(c) \leq f(x)$ " in the prior def.

Theorem (Fermat, ~1636) If f has a rel. max or min at $x=c$, then either $f'(c)$ is 0 or undefined.

NOTE: The reverse is not true



Def Let c be in the domain of f .
 If $f(c)$ is 0 or undefined, we say c is
 a critical number (critical point, critical value)
 of f .

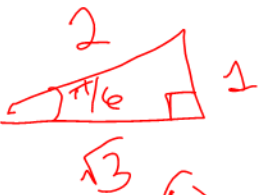
Example 1 Let $f(x) = 2 \cos(5x) + 5x$

@ Is $x = \frac{\pi}{30}$ a critical number of f ?

$$f'(x) = -10 \sin(5x) + 5$$

$$f'\left(\frac{\pi}{30}\right) = -10 \sin\left(5 \cdot \frac{\pi}{30}\right) + 5 = -10 \sin\left(\frac{\pi}{6}\right) + 5$$

$$= -10\left(\frac{1}{2}\right) + 5 = 0 \quad \checkmark \text{ Yes}$$



⑥ Is $x = \frac{\pi}{5}$ a crit. num. of f ?

$$f'(x) = -10 \sin(5x) + 5$$

$$f'\left(\frac{\pi}{5}\right) = -10 \sin\left(5 \cdot \frac{\pi}{5}\right) + 5 = 5 \neq 0 \text{ or undefined}$$

X not a crit. num.

Example 2 Find all crit. nums. of

$$f(x) = x^3 - 75x$$

$$f'(x) = 3x^2 - 75$$

$f'(x)$ undefined: Never

$$f'(x) = 0 : \quad 3x^2 - 75 \stackrel{?}{=} 0$$

$$3(x^2 - 25) = 0$$

$$3(x-5)(x+5) = 0$$

$$\Rightarrow \boxed{x = 5, -5}$$

Example 3 Find all crit. nums. for

$$f(x) = x^3 + 5x^2 + 7x$$

$$f'(x) = \underbrace{3}ax^2 + \underbrace{10}bx + \underbrace{7}c \quad \underline{\underline{\text{set } 0}}$$

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-10 \pm \sqrt{(10)^2 - 4(3)(7)}}{2(3)} = \frac{-10 \pm \sqrt{100 - 84}}{6}$$

$$= \frac{-10 \pm \sqrt{16}}{6} = \frac{-10 \pm 4}{6}$$

$$\Rightarrow x = \frac{-10+4}{6}, \frac{-10-4}{6} \Rightarrow x = -1, -\frac{7}{3}$$

Example 4 Repeat for $f(x) = 4x^3 - \frac{4}{x^3}$

$$f'(x) = 12x^2 - [4x^{-3}]' = 12x^2 + 12x^{-4}$$

$$= 12x^2 + \frac{12}{x^4}$$

NOTE: It is tempting to say 0 is a crit. num. here. But, remember crit. nums. need to be in the domain of f .

$f'(x)$ is never undefined (in the domain of f)

$$f'(x) = 0 :$$

$$12x^2 + \frac{12}{x^4} \stackrel{?}{=} 0$$

$$x^4 \left(12x^2 + \frac{12}{x^4} \right) = x^4 \cdot 0$$

$$12x^6 + 12 = 0$$

$$x^6 + 1 = 0$$

NO SOLUTION

$\Rightarrow f$ has no crit. numbers.

Example 5 Repeat for $f(x) = x e^x$

$$f'(x) = (1) e^x + x e^x$$

f' is never undefined, we need to solve

$$1 \cdot e^x + x e^x = 0$$

$$e^x (1 + x) = 0$$

$$e^x = 0$$

Never

OR

$$1 + x = 0$$

$$\boxed{x = -1}$$

Ex 6 $f(x) = 6x^4 e^{5x+4}$

$$f'(x) = 24x^3 e^{5x+4} + 6x^4 [5e^{5x+4}]$$
$$= 24x^3 e^{5x+4} + 30x^4 e^{5x+4} \quad \underline{\underline{\text{set } 0}}$$

$$6x^3 e^{5x+4} [4 + 5x] = 0$$

$$6x^3 = 0 \quad \text{OR} \quad \underbrace{e^{5x+4}}_{\text{Never}} = 0 \quad \text{OR} \quad 4 + 5x = 0$$
$$\boxed{x = 0} \quad \boxed{x = -\frac{4}{5}}$$

Ex 7 Repeat for $f(x) = \frac{\ln x}{x}$

$$f'(x) = \frac{\frac{1}{x} \cdot x - (1) \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$f'(x)$ is never undefined (in the domain of f)

$$\frac{1 - \ln x}{x^2} = 0$$

$$1 - \ln x = 0$$

$$\ln x = 1$$

$$\boxed{x = e}$$

